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ORIGINAL PAPER

Sequential Indentation Tests to Investigate the Influence of Confining Stress on Rock Breakage by Tunnel Boring Machine Cutter in a Biaxial State

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Abstract The influence of confining stress on rock breakage by a tunnel boring machine cutter was investigated by conducting sequential indentation tests in a biaxial state. Combined with morphology measurements of breaking grooves and an analysis of surface and internal crack propagation between nicks, the effects of maximum confining stress and minimum stress on indentation efficiency, crack propagation and chip formation were investigated. Indentation tests and morphology measurements show that increasing a maximum confining stress will result in increased consumed energy in indentations, enlarged groove volumes and promoted indentation efficiency when the corresponding minimum confining stress is fixed. The energy consumed in indentations will increase with increase in minimum confining stress, however, because of the decreased groove volumes as the minimum confining stress increases, the efficiency will decrease. Observations of surface crack propagation show that more intensive fractures will be induced as the maximum confining stress increases, whereas the opposite occurs for an increase of minimum confining stress. An observation of the middle section, cracks and chips shows that as the maximum confining stress increases, chips tend to form in deeper parts when the minimum confining stress is fixed, whereas they tend to formed in shallower parts as the minimum confining stress increases when the maximum confining stress is fixed.

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Keywords Confining stress · TBM cutter · Biaxial state · Breaking groove morphology · Indentation efficiency

1 Introduction

As tunnel boring machine (TBM) excavation depths increase, high confinement has proven to be a critical factor that influences cutting efficiency (Yin et al. 2014a, b). Beneficial and adverse influences exerted by the in situ stress on cutting efficiency have been reported. Because of poor rock breakage at the working face and an increased consumed TBM power, Gong's study (2010) showed that the cutting efficiency will be decreased by high in situ stresses, which were close to 30 MPa in the Jing Ping hydropower station in China. However, it has been reported that rock breakage will increase significantly by fractures caused by high in situ stress at the working face (Tarkoy and Marconi 1991). Innaurato's indentation tests showed that the chipping mechanism improved as soon as the overburden increased (Innaurato et al. 2007). Furthermore, according to measurements of effective crack length, the numerical study by Ma proposed that rock fragmentation was promoted when the confining stress was higher than a critical value, and vice versa (Ma et al. 2011). The above research indicates that rock fragmentation, which is influenced by confining stress, is critical to the cutting efficiency. Chipping between cutters has proven to be efficient in rock fragmentation (Gertsch et al. 2007; Cho et al. 2013). With respect to the chipping mechanism, two popular patterns, which are the shear failure and the tensile fracture pattern, have been proposed on the chip formation between cutters (Ozdemir et al. 1979; Gertsch 2000). Previous studies show that besides a confining stress, many factors, such as discontinuities in rocks and TBM cutter

loading modes and configurations, would exert a significant influence on chip formation and TBM performance (Bejari and Khademi Hamidi 2013; Gong et al. 2005, 2006a, b; Carpinteri and Invernizzi 2005; Lindqvist 1984; Wang et al. 2011; Moon and Oh 2012; Ma et al. 2011; Huang et al. 1998; Innaurato et al. 2007; Farrokh et al. 2012). In most, if not all cases, the above studies have been conducted in a plane condition. These researches contribute significantly to an understanding of chipping formation in the plane parallel to the plane determined by adjacent cutters' axes along the indentation direction. However, chips, as fragments with several facets, will be formed completely not only by crack aggregation in deeper internal rocks, but also by the incision of surface cracks initiated from the rock surface perpendicular to penetration. However, no investigations of these surface cracks can be conducted in most numerical and experimental studies conducted on two-dimensional models (Ma et al. 2011; Innaurato et al. 2011; Wang et al. 2011; Chen and Labuz 2006; Gong et al. 2006b; Liu et al. 2002). In most indentation tests, which are viewed as the main breakage process for rock fragments by TBM cutters (Yin et al. 2014a; Hood and Roxborough 1989), only one indentation process was conducted, and the surface cracks between adjacent cutting grooves cannot be investigated simultaneously (Yin et al. 2014a; Innaurato et al. 2007; Chen and Labuz 2006). Minimal attention has been given to these surface cracks. Until recently, Liu's and Zhang's investigations found that these cracks may be critical to chip formation and cutting efficiency (Liu et al. 2015; Zhang et al. 2012).

In this work, indentation tests were conducted on granite and sandstone specimens in a biaxial state to investigate the influence of confining stress on rock breakage. The groove morphologies for varying confining stress levels between cutters were measured using a Talysurf CLI 2000 high-resolution three-dimensional surface profiling system (Taylor Hobson, Leicester, England). An analysis of the results obtained from indentation tests and morphology measurements of the grooves yields cutting efficiencies for different confining stress levels. An observation of the surface and internal crack distribution has also been conducted.

2 Laboratory Tests

2.1 Test Apparatus

To conduct indentation tests of TBM cutters under nearly biaxial stress conditions, as shown in Fig. 1a, we applied a triaxial testing machine that was designed by Central South University. The maximum horizontal and vertical thrust forces, which reach up to 2000 and 3000 kN, respectively, meet the requirements of confinement and indentation force provision. The loading frames of this machine are made of high stiffness materials, and during indentation, the TBM cutter can be displacement- or force-controlled and the indentation depth, indentation force and confinement can be recorded and restored. As shown in Fig. 1b, the TBM cutter used is shaped as a portion of the TBM cutter ring. The cutter is made of heat-treated high hardness steel, and its detailed geometry is shown in Fig. 2. Compared with the cutter cross section in the investigation by Rostami (2013), the cutter shape proved to be reasonable. In this work, an acoustic emission (AE) system is used to monitor the AE events in the indentation tests. As shown in Fig. 1b, the AE sensor is glued on the top surface of the rock specimens, and the trigger level is set to 40 dB. The morphology-measuring machine, Talysurf CLI 2000, is shown in Fig. 3. As shown in Fig. 3, the rock specimens can be placed on the loading plate after the indentation tests. The morphology of the specific area will be obtained by conducting origin setting and scanning, in sequence.

2.2 Rock Sample

To investigate the influence of confining stress on the fracture conditions in different rocks, two types of rock samples were used. The first type is a granite specimen that contains quartz, feldspar and biotite. Its grain size ranges from 0.4 to 4 mm. The other is a sandstone specimen with bedding planes, with a bedding distance of almost 1 mm. All specimens were incised from a big granite or sandstone block, so their mechanical properties are similar. The bedding plane in the indentation tests is parallel to the plane determined by Points A, B and C in Fig. 4. The main mechanical parameters as listed in Table 1 were obtained from uniaxial, Brazilian tensile and direct shear tests according to ISRM recommendations. According to the classification of rock brittleness proposed by Yagiz (2009), the granite and sandstone specimens used can be classified as moderately brittle and ductile rocks, respectively.

The loading capacity of Talysurf CLI 2000 is 15 kg, so, combined with the rock density specimens, the maximized specimen sizes are 25, 20 and 10 in length, width and height, respectively. According to previous analysis (Yin et al. 2014b; Johnson 1985), the size effect can be eliminated if the ratio of the specimen width to plastic zone depth of a plastic zone is higher than 6. Based on the cavity model developed by Alehoseein et al. 2000), the plastic zone depths of the granite and sandstone specimens are 1.9 and 2. 0 mm. The minimum width of the specimen is 100 mm, and the corresponding minimum ratios are nearly 52 and 50, which are much higher than 6. So, it can be concluded that the specimen size is reasonable.

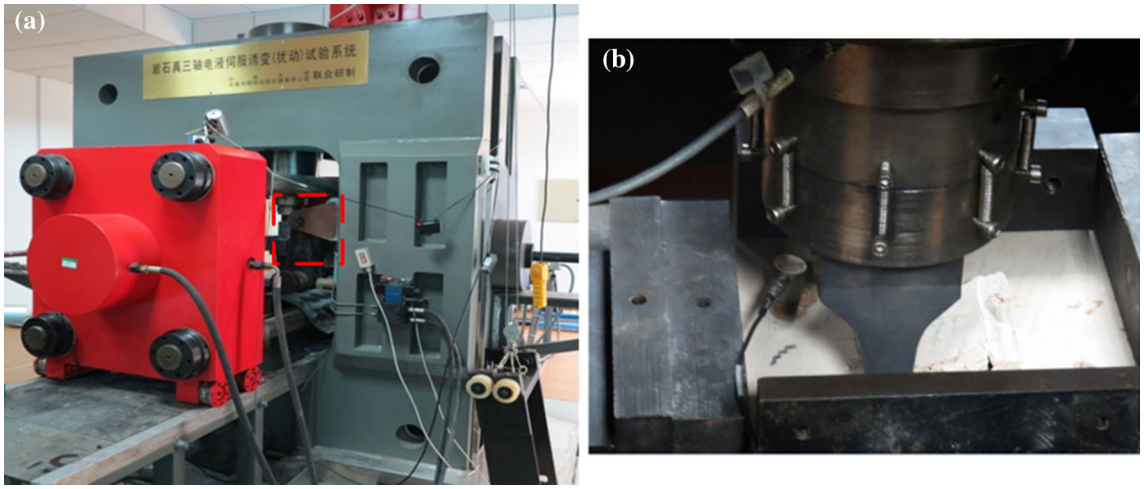


Fig. 1 Test equipment for indentation tests, **a** picture of triaxial testing machine; **b** magnified picture of the rectangle area

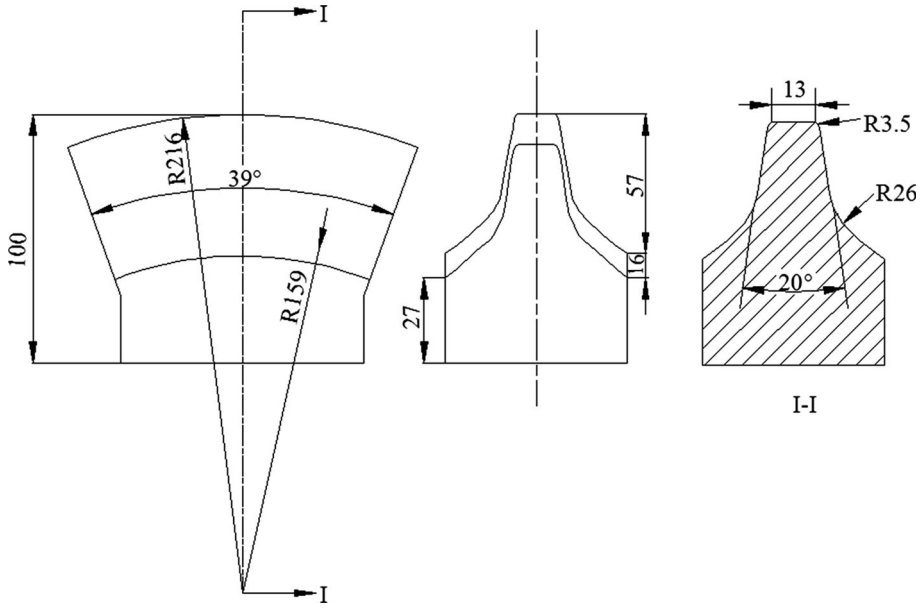


Fig. 2 Sketch of the TBM indentation cutter

2.3 Testing Design and Procedures

The red and green arrows in Fig. 4 denote the direction in which the maximum and minimum confining stresses act, respectively. Considering the difference in strength between granite and sandstone specimens, the minimum confining stresses exerted on granite specimens are 5, 10 and 15 MPa, with corresponding maximum confining stresses of 5, 10, 15, 20 and 25 MPa. The minimum confining stresses applied to sandstone specimens are 5, 7.5 and 10 MPa, and the corresponding maximum confining stresses are 5, 7.5, 10, 12.5 and 15 MPa. A phase angle exists between installed adjacent TBM cutters, so, the rock breakage between cutters can be viewed as rock breakage induced by indentations in sequence. As shown in Fig. 4, the spacing between cutters is kept at 70 mm. In indentation tests, the cutter is displacement-controlled. The final indentation depth that is deep enough to cause fracturing between grooves is kept at 6 mm based on a study conducted by Gertsch (Gertsch et al. 2007). An acoustic emission (AE) sensor was placed at the rock surface to monitor the AE hits in the indentation process. Crack observation in the plane determined by Points A, B and C and morphology measurements of the groove were also conducted with detailed procedures as follows:

1. To eliminate the friction effect, lubricate the specimen surface and the loading plates before placing the specimens in a designated place.

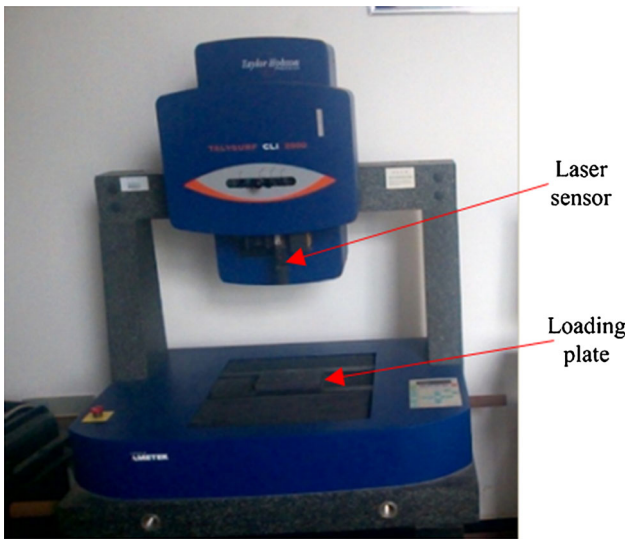


Fig. 3 Morphology measuring equipment

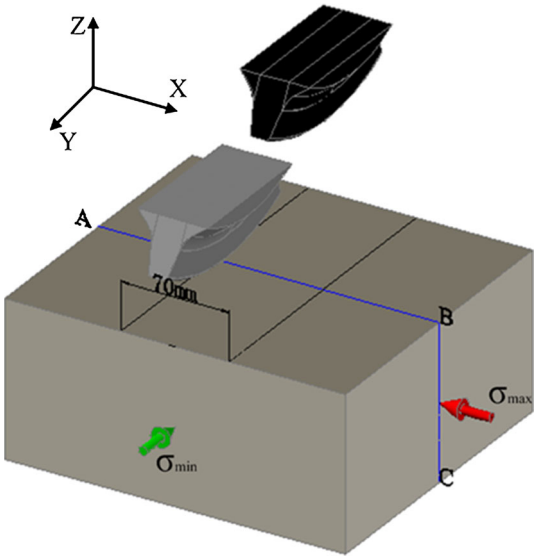


Fig. 4 Schematic diagram of indentation test

Table 1 Mechanical properties of rock samples

Mechanical properties	Granite	Sandstone
Elastic modulus (GPa)	22.3	17.1
Poisson's ratio	0.23	0.25
Uniaxial compressive strength (MPa)	94.8	29.5
Tensile strength (MPa)	7.8	9.2
Friction angle (°)	49.2	41.6
Density (g/cm^3)	2.9	2.5

2. Slowly apply confining forces of 10 kN in both confinement directions; as the specimens reach stability, exert the confining stress to proportional specific values. In this work, the stress level is represented by $\sigma_{min} - \sigma_{max}$. For example, 5–10 MPa denotes that the minimum confining stress equals 5 MPa, whereas the maximum confining stress is 10 MPa.
3. Install the AE sensor on the specimen, and place the cutter in a specific location. After installment of the AE sensor and TBM cutter, apply a load that is 100 N on the TBM cutter to ensure sufficient contact between the cutter and loading cell. Then exert a displacement-controlled load of 0.5 mm/min on the cutter. When the indentation depth reaches 6 mm, stop loading.
4. Unload the TBM cutter thrust, and repeat Step 3.
5. After the indentation test, place the specimen on the loading plate of Talysurf CLI 2000, set the origin and start scanning. As shown in Fig. 5, the scanning area is determined by the outline of chips and the centerline of nicks.
6. Cut the specimen along the plane determined by Points A, B and C (shown in Fig. 4) to investigate crack distributions.

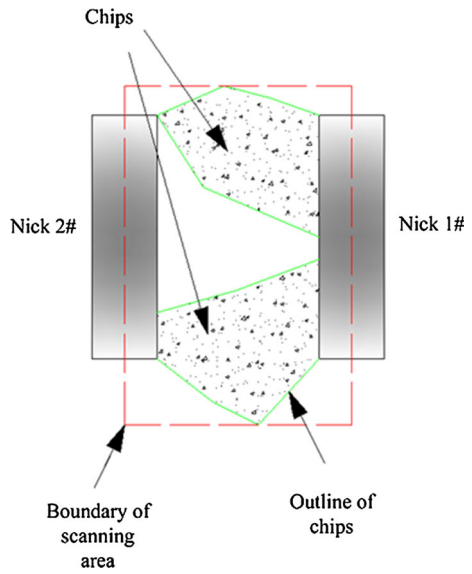


Fig. 5 Scanning area

3 Test Results and Analysis

3.1 Indentation Forces and AE Events in the Indentation Process

Figure 6 shows the typical indentation force and AE events in the first indentation process conducted on the granite specimen. According to the characteristics of indentation force and AE events, the first indentation process is divided into three stages. In Stage A, the AE events remain low,

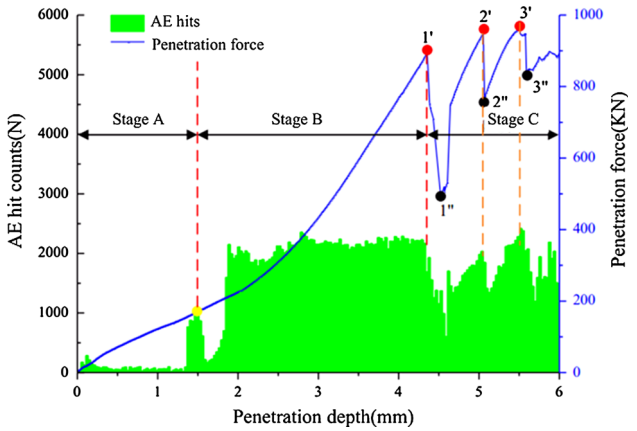


Fig. 6 Typical indentation forces and AE events in the first indentation process

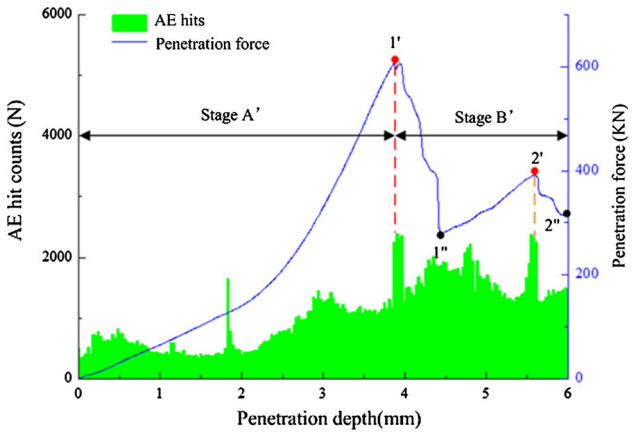


Fig. 7 Typical indentation forces and AE events in the second indentation process

and the indentation force increases relatively slowly as the indentation depth increases. As the depth reaches a critical value, which is nearly 1.5 mm at 10–25 MPa, the recorded AE events in unit time reach up to 1000, and the corresponding indentation force is close to 200 kN. As shown in Fig. 6, the indentation force increases rapidly in Stage B, and the AE events in unit time remain at a relatively high level. As shown in Fig. 6, the first peak indentation force (Point 1′) is encountered when the depth is 4.42 mm and the corresponding peak force reaches up to 900 kN. As the indentation depth increases further, the indentation process moves to Stage C. In Stage C, fluctuations of indentation forces reported in previous studies are observed (Innaurato et al. 2007; Innaurato et al. 2011; Chen and Labuz 2006). As shown in Fig. 6, peak indentation forces are observed when the depths are 4.42, 5.05 and 5.51 mm, and the corresponding depths as the indentation force reaches the bottom points are 4.50, 5.08 and 5.58 mm, respectively. The orange and red dashed lines indicate that corresponding AE events at peak points remain at a relatively high level; however, as the indentation proceeds, recorded AE events and the indentation force drop significantly between the corresponding peak and bottom points. As the depth increases from a bottom to a later peak point, the corresponding indentation force and AE events will increase rapidly. These AE characteristics are similar to those in the study by Entacher (Entacher et al. 2014).

The corresponding typical indentation force and AE events in the second indentation process are shown in Fig. 7. No critical indentation depth exists in the indentation process. AE events stay at a high level throughout the indentation process; thus, according to characteristics of

Table 2 Average peak indentation force and consumed energy in different confining stresses

	Stress level	Average peak indentation force (KN)			Total consumed energy (J)		
Granite	$\sigma_{min}/\sigma_{max}$	5 MPa	10 MPa	15 MPa	5 MPa	10 MPa	15 MPa
	5 MPa	544			2583		
	10 MPa	686	746		3471	3268	
	15 MPa	718	617	784	3139	3169	3224
	20 MPa	784	718	969	3310	3425	4941
	25 MPa	797	786	941	3948	4314	5268
Sandstone	$\sigma_{min}/\sigma_{max}$	5 MPa	7.5 MPa	10 MPa	5 MPa	7.5 MPa	10 MPa
	5 MPa	459			2644		
	7.5 MPa	561	511		2952	3161	
	10 MPa	583	642	617	3779	4137	4207
	12.5 MPa	627	707	626	3747	4182	4415
	15 MPa	654	717	694	4197	4308	4673

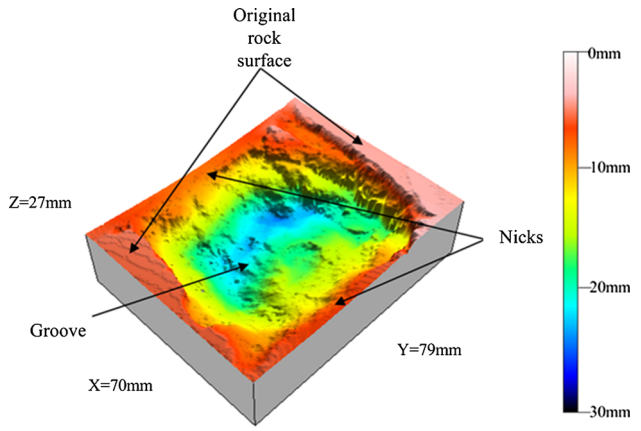


Fig. 8 Typical groove morphology

the indentation force, the indentation process is only divided into two stages, A' and B'. The characteristics in Stages A' and B' are similar to those in Stages B and C in the first indentation process.

Corresponding peak forces and energy consumed in indentations conducted at different confining stresses, and which are important indices that characterize the indentation efficiency in the indentation processes (Bejari and Khademi Hamidi 2013; Gertsch et al. 2007; Cho et al. 2013), are listed in Table 2.

3.2 Groove Morphology

The groove volume between nicks is another critical index that determines the cutting efficiency of TBM cutter operation (Gertsch et al. 2007; Cho et al. 2013). The groove morphology may shed some light on the chip formation mechanism between nicks to some extent; therefore, grooves formed between nicks were investigated based on morphology measurements conducted on the Talysurf CLI 2000. According to the coordinated system set in Fig. 4, a typical groove morphology for the granite specimen when the confining stress is 10–20 MPa is shown in Fig. 8. The gauge indicates that the depth of the groove surface increases as the gauge color evolves from white to blue. Combined with the scanning area shown in Fig. 5, it can be determined that the groove is surrounded by nicks and intact rock surfaces, and that the typical groove is nearly 'basin'-shaped. As shown in Fig. 8, the uneven groove surfaces may shed some light on the chipping

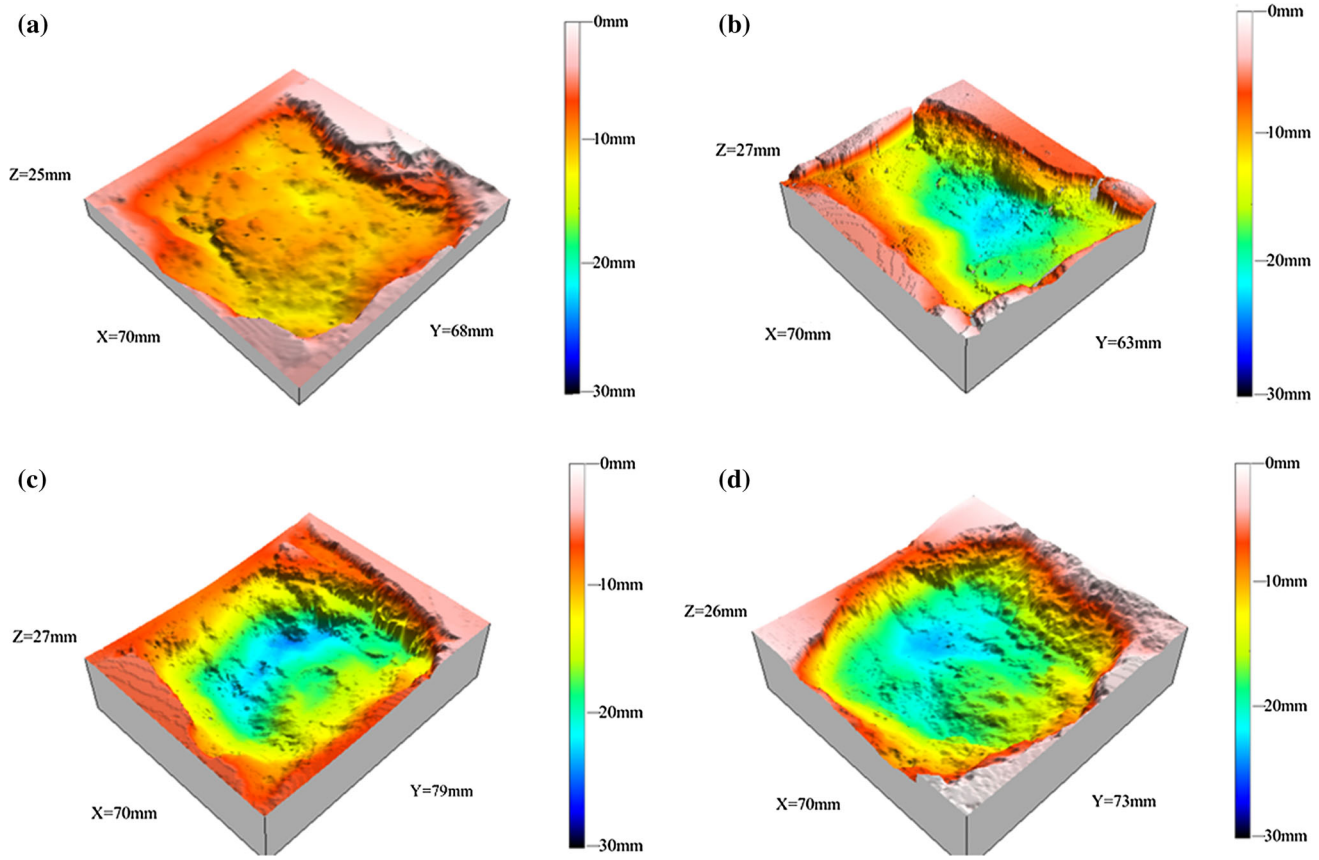


Fig. 9 Groove morphologies as the minimum confining stress is 10 MPa: **a**–**d** are the groove morphologies when the maximum confining stresses are 10, 15, 20 and 25 MPa, respectively

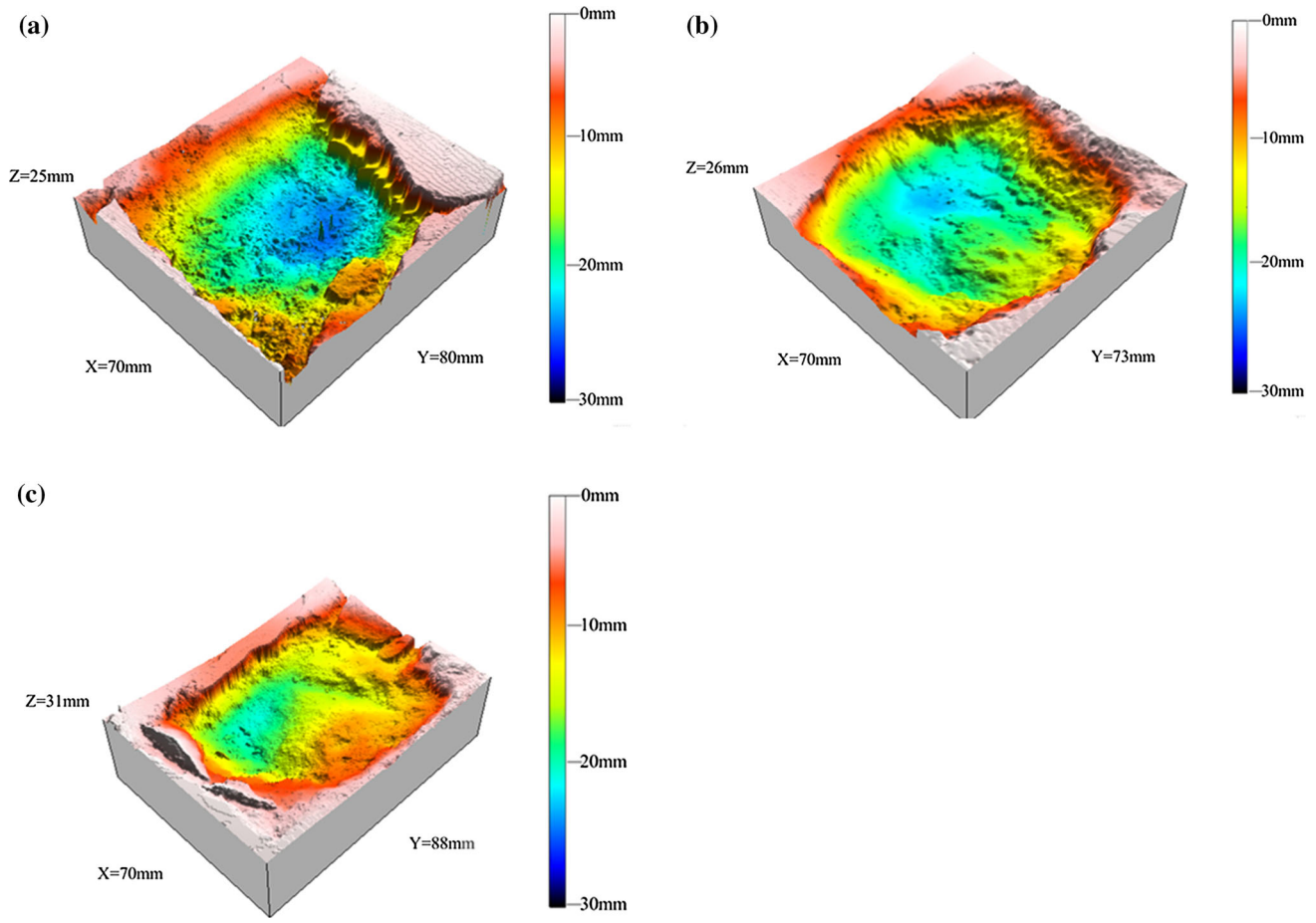


Fig. 10 Groove morphologies as the maximum confining stress is 25 MPa: **a–c** are the groove morphologies when the minimum confining stresses are 5, 10 and 15 MPa, respectively

Table 3 Groove volumes of granite and sandstone specimens subjected to varying confining stresses

Groove volumes in granite (mm^3)				Groove volumes in sandstone (mm^3)			
$\sigma_{min}/\sigma_{max}$	5 MPa	10 MPa	15 MPa	$\sigma_{min}/\sigma_{max}$	5 MPa	7.5 MPa	10 MPa
5 MPa	25634			5 MPa	5358		
10 MPa	33682	28333		7.5 MPa	14287	32343	
15 MPa	61370	47705	38753	10 MPa	37492	40114	21262
20 MPa	58768	51687	52609	12.5 MPa	75434	46835	40412
25 MPa	87031	73908	65388	15 MPa	68259	58169	62163

mechanism between nicks. To investigate the effects of confining stress on the groove, the corresponding groove morphologies of granite and sandstone specimens are shown in Figs. 9 and 10.

To investigate the effects of maximum confining stress on groove morphology of granite specimens, for example, the groove morphologies of granite specimens subjected to increasing maximum confining stresses when the minimum confining stress is 10 MPa, are presented in Fig. 9. Figure 9a indicates that an island exists at the center of the groove, that the maximum depth is 12.4 mm and that the total groove volume is 28,333 mm^3. Compared with the typical groove morphology shown in Fig. 8, the groove in this stress level indicates that it tends to be formed in shallow parts of the specimens. As the maximum confining stress increases to 15 MPa, as shown in Fig. 9b, the maximum depth and corresponding volume of the nearly 'basin'-shaped groove increase to 21.9 mm and 47705 mm^3,

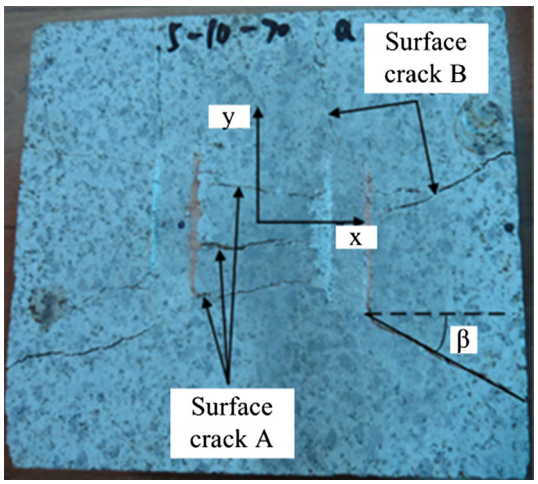


Fig. 11 Typical surface cracks

respectively, however, the fracture scope in the Y direction decreases slightly to 63 mm. Figure 9c shows that the maximum depth and groove volume increases to 25.1 mm and 51,687 mm^3, respectively, when the maximum confining stress increases to 20 MPa. The increased groove volume results from an increased fracture scope in the Y direction and an increased proportion of deeper to total grooves. As shown in Fig. 9d, the maximum depth decreases slightly to 24.2 mm, however, because the proportion mentioned above increases further, the groove volume increases to 73,908 mm^3. It is not difficult to determine from the above descriptions that as the maximum confining stress increases, grooves tend to form in deeper rock specimens, and as shown in Table 3 that this increased groove depth will result in an increase in groove volume.

To investigate the effects of minimum confining stress on groove morphology of granite specimens, for example, Fig. 10 depicts the groove morphologies of granite specimens subjected to increasing minimum confining stresses when the maximum confining stress is 25 MPa. The groove surface in Fig. 10a is similar to that in Fig. 9d, but the maximum depth increases to 27.1 mm, the proportion of the groove that is deeper than 18 mm remains high, and the corresponding groove volume is high up to 87,031 mm^3. As the minimum confining stress increases to 10 MPa, the maximum groove depth drops slightly to 24.8 mm, and the measured groove volume decreases significantly to 73,908 mm^3. The decreased proportion of deep groove parts, decreased maximum depth and fracture scope in the Y direction are responsible for the decrease in groove volume. As shown in Fig. 10c, the groove seems to be formed at shallow parts of the rock specimens since the minimum confining stress increases to 15 MPa. The maximum groove depth drops slightly to 24.5 mm, but it can be observed easily that the proportion of deep grooves decreases significantly. The corresponding groove volume decreases significantly to 65,388 mm^3. It can be concluded from the above analysis that the increase in minimum confining stress will result in a decrease in groove volume between nicks. The decreased volume may be induced by the decreased groove depth as the minimum confining stress increases.

3.3 Surface Cracks Between Nicks

Based on the crack locations, two types of surface cracks are visible in Fig. 11. The first type of surface cracks, which are termed Surface cracks A, are located between nicks. The others, which are termed Surface cracks B, are cracks that initiate from the rims of the nick and propagate to the rims of the rock sample. The angle determined by the

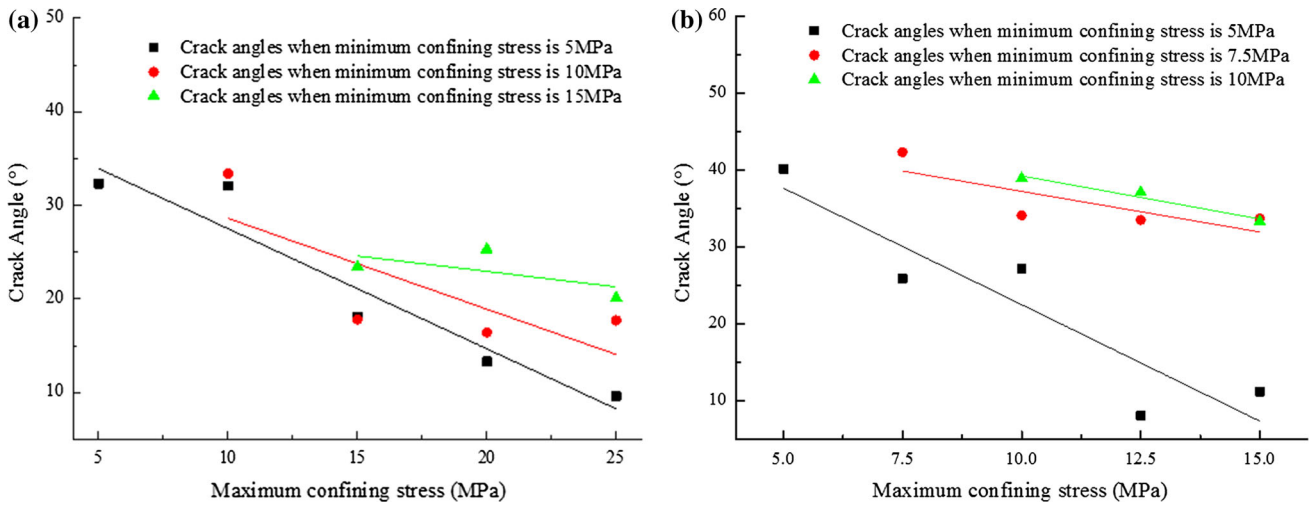


Fig. 12 Surface crack angles in varying confining stress levels: **a**, **b** are the surface crack angles of granite and sandstone specimens in varying confining stress levels, respectively

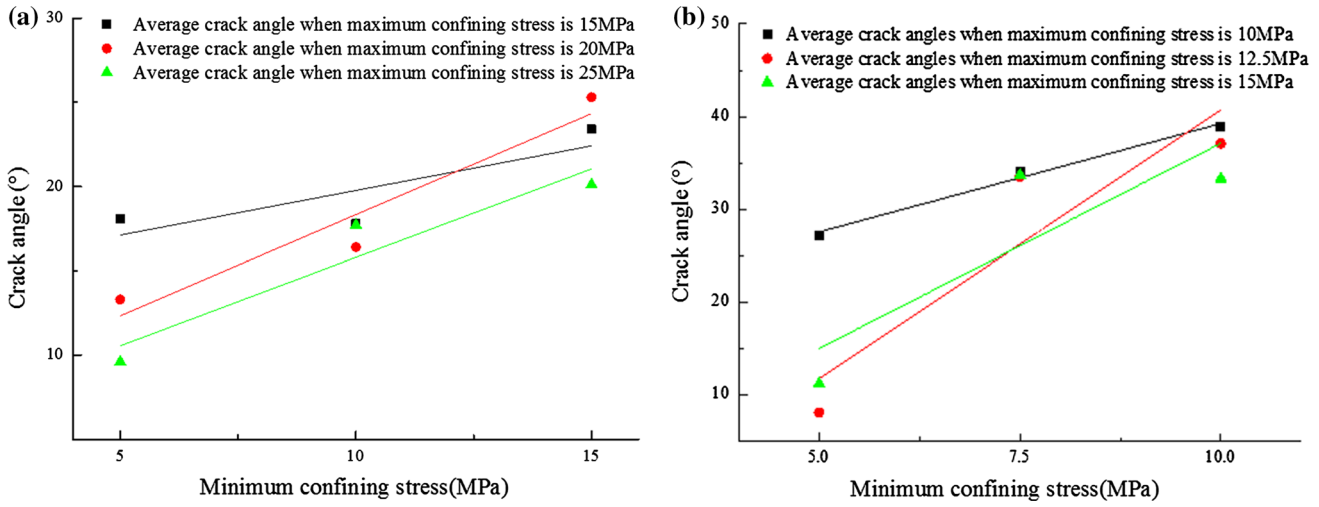


Fig. 13 Surface crack angles in varying confining stress levels: **a**, **b** are the surface crack angles of granite and sandstone specimens in varying confining stress levels, respectively

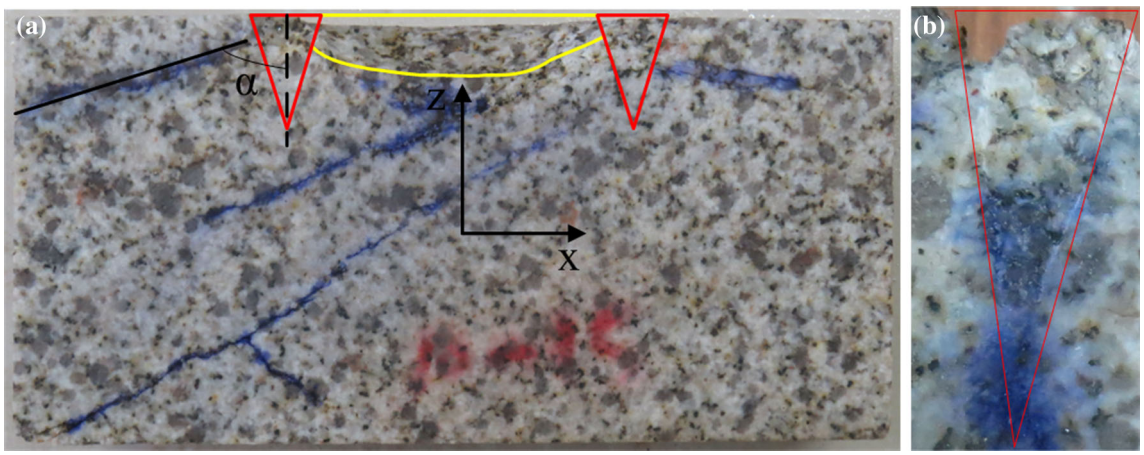


Fig. 14 Typical fracture condition and plastic zones in the plane: **a** fracture condition; **b** typical plastic zones

propagation direction of the surface cracks and the X axis shown in Fig. 11 is defined as the surface crack angle β. Combined with the introduction of chip formation above, it can be inferred from Fig. 11 that the propagation of Surface crack A is critical to rock breakage between nicks. So, to investigate the influence of confining stress on the propagation characteristics of surface cracks, the measured crack angles are shown in Figs. 12 and 13.

3.3.1 Surface Crack Distributions as Maximum Confining Stress Increases

The measured surface crack angles are shown in Fig. 12. As shown in Fig. 12a, the average surface crack angle reaches 32.3° for 5–5 MPa. As the maximum confining stress increases to 10 MPa, more Surface cracks A are observed and the corresponding crack angle decreases slightly to 32.1°. When the maximum confining stress reaches 15 MPa, aggressive rock breakage between nicks is observed, and the corresponding surface crack angle decreases significantly to 18.1°. As the maximum confining stress increases to 20 and 25 MPa, increasingly intensive chippings are observed, and the crack angles tend to decrease further. It can be concluded from the above descriptions that increasingly dense Surface cracks A and decreasing average crack angles result from the increasing maximum confining stress when the minimum confining stress is 5 MPa in the granite specimens. At other minimum stress levels, a similar tendency of crack angles is observed. Figure 12b shows that as the maximum

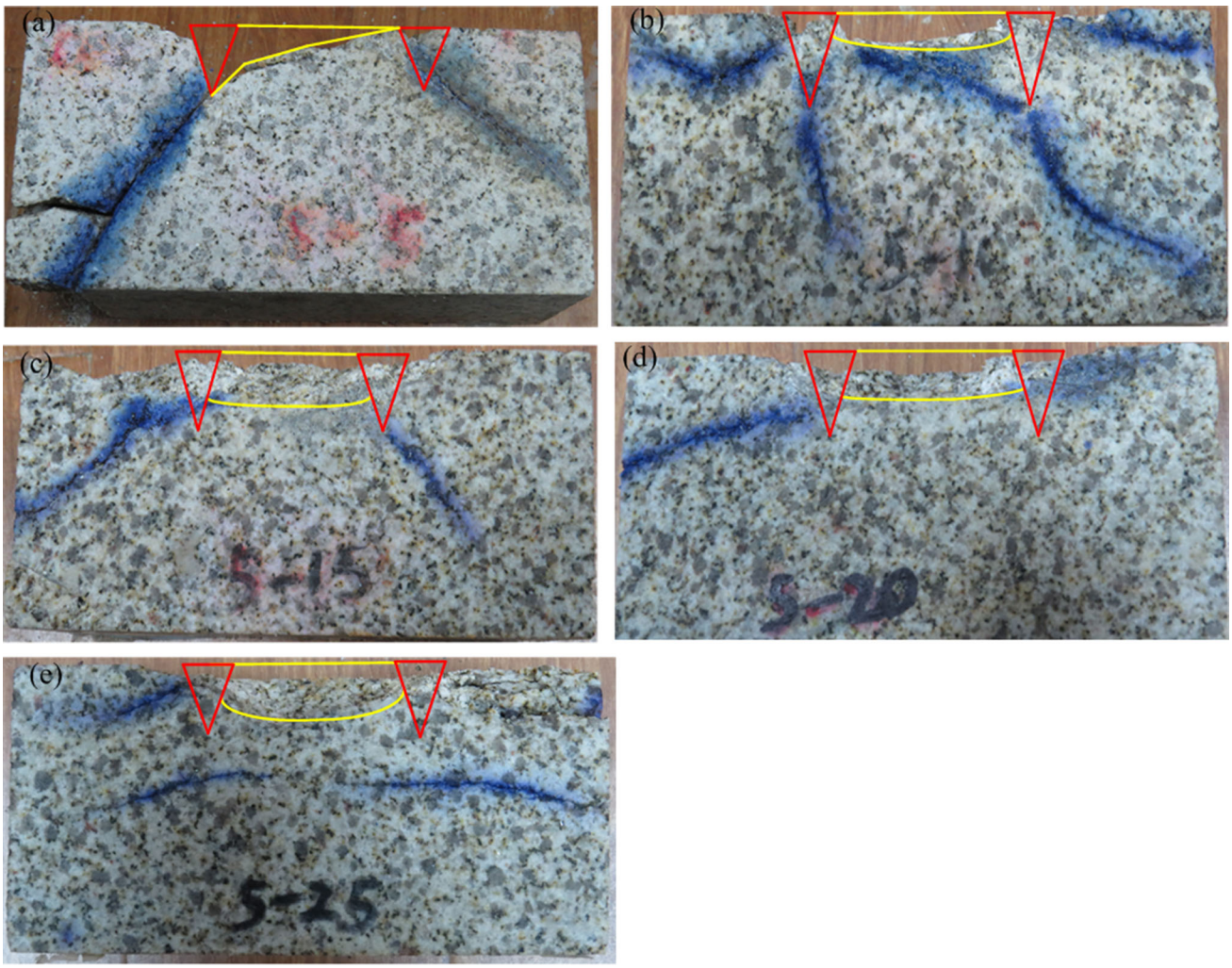


Fig. 15 Internal crack distributions of granite specimens when the minimum confining stress is 5 MPa: **a**–**e** are the internal crack distributions when maximum confining stress are 5, 10, 15, 20 and 25 MPa, respectively

confining stress increases, the decreases in surface crack angles are caused at specific minimum stress levels too.

3.3.2 Surface Crack Distribution as Minimum Confining Stress Increases

Figure 13 depicts the surface crack angles as the minimum confining stress increases. Figure 13a shows that the average crack angle is only 13.3° in granite specimens when the stress level is 5–20 MPa. As the minimum confining stress increases to 10 MPa, the number of Surface cracks A decreases and the corresponding crack angle increases slightly to 16.4°. When the minimum confining stress increases to 15 MPa, fewer surface cracks are visible between nicks and, as shown in Fig. 13a, the corresponding average surface crack angle increases to 25.3°. Combined with other curves shown in Fig. 13, it can be seen that the increase in minimum confining stress will result in an increase in surface crack angle.

3.4 Internal Cracks and Typical Chips Formed Between Nicks

As mentioned above, the crack distribution in internal rock specimens is another critical factor that affects chipping between nicks. Typical fracture conditions in internal rock are shown in Fig. 14a. Yellow lines denote the outline of the groove in this section. As shown in Fig. 14b, the typical plastic zone is nearly triangle-shaped, so, two red triangles in Fig. 14a represent adjacent plastic zones. The combination of cracks that propagate from adjacent plastic zones is visible in Fig. 14a. The outline of the groove and connected internal cracks overlap with each other in some segments. In this work, the angle between the Z and crack propagation directions is defined as the deflection angle of the internal cracks, α. To investigate the influence exerted by confining stress on the internal fracture, for example, fracture conditions of granite specimens when the minimum confining stress is 5 MPa are used to investigate the

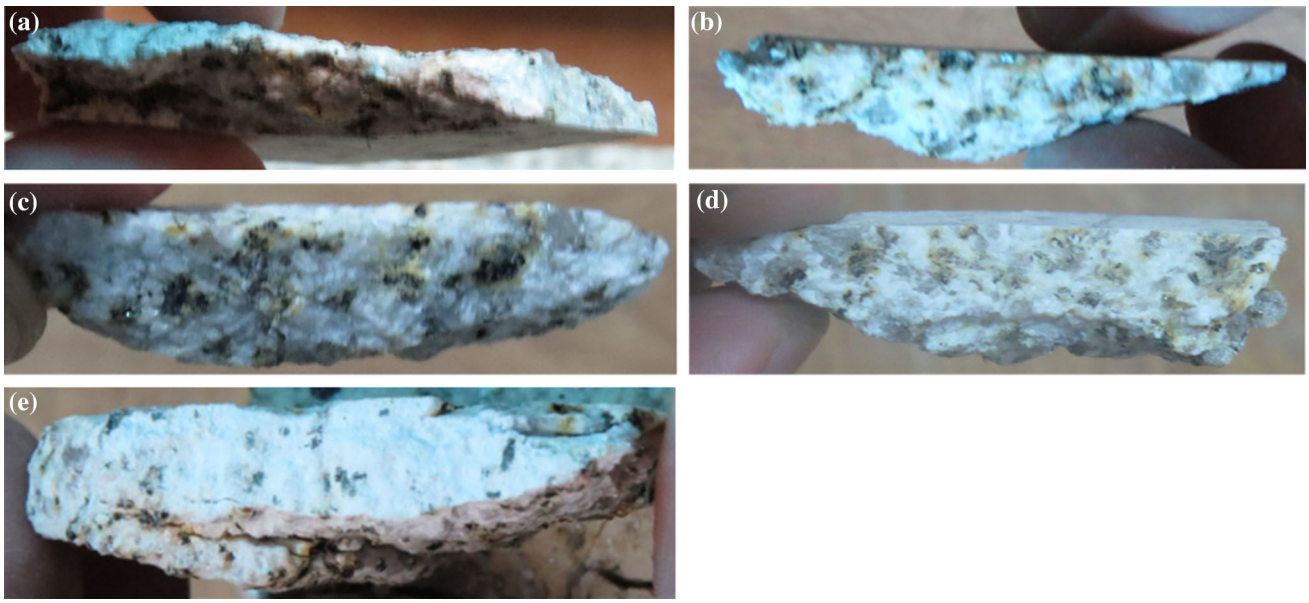


Fig. 16 Typical chips when the minimum confining stress is 5 MPa: **a–e** are the typical chips when maximum confining stress are 5, 10, 15, 20 and 25 MPa, respectively

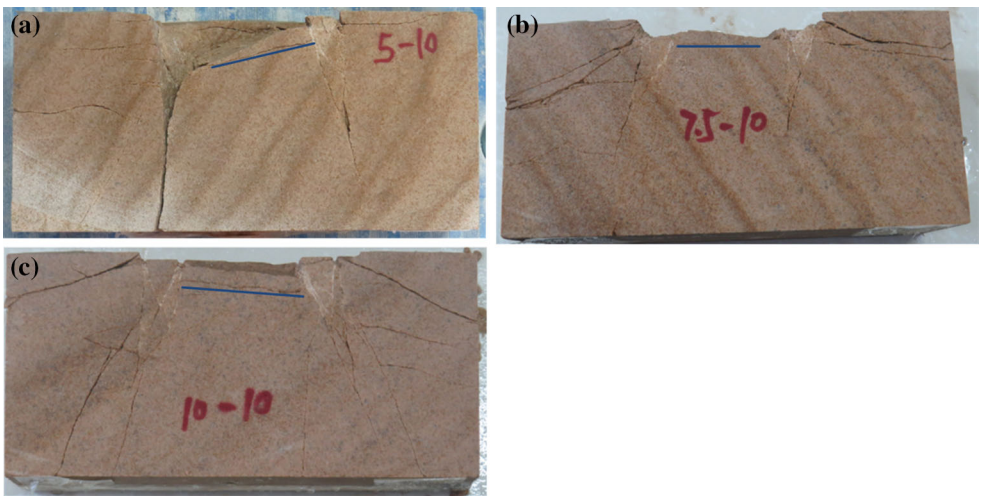


Fig. 17 Internal crack distributions of sandstone specimens when the maximum confining stress is 10 MPa: **a–c** are the surface crack distribution when minimum confining stress are 5, 10 and 15 MPa, respectively

influence exerted by maximum confining stress. Fracture conditions of sandstone specimens when the maximum confining stress is 10 MPa are used to investigate the effects exerted by minimum confining stress.

3.4.1 Internal Cracks Between Nicks and Typical Chips as Maximum Confining Stress Increases

Figure 15 depicts the internal crack distributions as the confining stress increases from 5 to 25 MPa when the minimum confining stress is 5 MPa. In the cutting process along the plane determined by ABC (shown in Fig. 4), the plastic zone may be flushed away by the cooling water of the cutting machine, so, as shown in Fig. 15a, one of the plastic zones is destroyed. Combined with the typical chips formed between nicks (shown in Fig. 16a), it can be inferred that the groove is formed at shallow parts of the rock specimen. Furthermore, the flat outlines of the groove and typical chip may indicate that chips may form by the combined action of shear failure and tensile fracture by the propagation of Surface crack A. The minimum deflection angle is 35.2° at this stress level. As the maximum confining stress increases to 10 MPa, Fig. 15b shows clearly that adjacent nicks are connected by cracks, and that the



Fig. 18 Typical chips when the maximum confining stress is 10 MPa: **a–c** are the typical chips when minimum confining stresses are 5, 7.5 and 10 MPa, respectively

outlines of the groove and the typical chip (shown in Fig. 16b) vary from the former. It can be seen from Fig. 15b that a partial overlap between the outline and the cracks between grooves is visible. The typical chip in this stress level shows clearly that half of the outline is uneven, whereas the rest is smooth. Combined with the formation mechanism of grooves, it can be determined that the chip may be formed by the combined action of the development of Surface crack A, the shear failure and the internal crack propagation between nicks. As the maximum confining stresses increase to 15, 20 and 25 MPa, the grooves formed between nicks are similar to those when the stress level is 5–10 MPa, and they tend to be deeper in middle parts. The minimum deflection angle increases significantly to 42.8°, 66.2° and 68.3°. The increase in minimum deflection angle indicates that cracks tend to propagate to the horizontal directions as the maximum confining stress increases. This tendency agrees with previous studies (Ma et al. 2011; Yin et al. 2014a; Innaurato et al. 2007). The typical chips in Fig. 16 show that chips are thicker in the middle, and this shared characteristic matches groove geometries perfectly. Uneven outlines of these chips and grooves indicate that chips in these stress levels may form by the coaction of propagation of Surface crack A and the internal cracks between grooves. It can be seen in Fig. 15e that two horizontal cracks are visible in internal rock specimens, and that these cracks are induced by high confining stress as reported in previous studies (Tarkoy and Marconi 1991; Yin et al. 2014a). The above results of internal crack propagation resulted from an increasing confining stress and show that increasing maximum confining stress may exert an effect on optimum spacing between cutters.

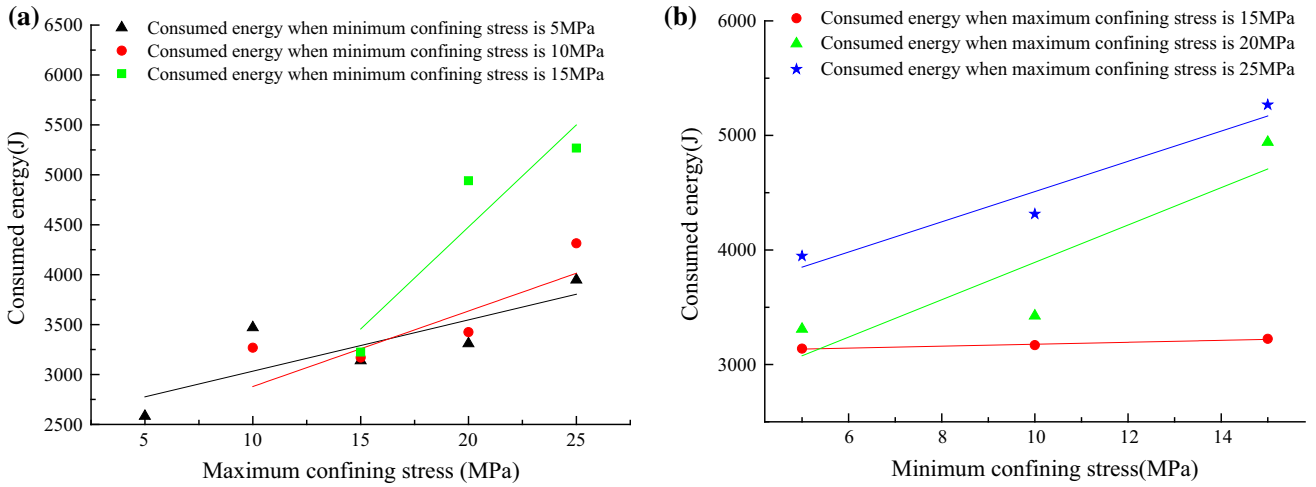


Fig. 19 Consumed energy in the indentation process conducted on granite specimens: **a** consumed energy as the maximum confining stress increases; **b** consumed energy as the minimum confining stress increases

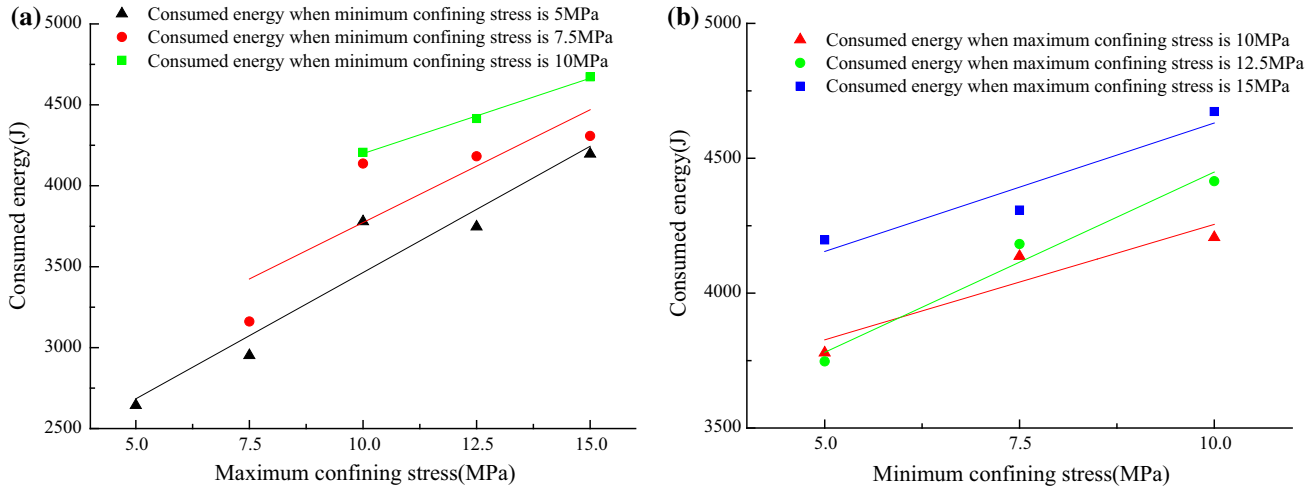


Fig. 20 Consumed energy in the indentation process conducted on sandstone specimens: **a** consumed energy as the maximum confining stress increases; **b** consumed energy as the minimum confining stress increases

3.4.2 Internal Cracks Between Nicks and Typical Chips as Minimum Confining Stress Increases

Figure 17 shows the fracture conditions of sandstones when the maximum confining stress is 10 MPa. In these sections, triangle-shaped plastic zones are visible. Plastic zones are connected by internal cracks denoted by blue lines in corresponding sections. As shown in Fig. 17a, a partial overlap of the groove and internal crack outlines is visible. The lateral view of a corresponding typical chip shown in Fig. 18a matches the groove outline perfectly. According to the above analysis, chips in this stress level may be formed by shear and tensile failure between nicks. However, as the minimum confining stress increases to 7.5 MPa, as shown in Fig. 17b, the groove is formed in a shallow part of the specimen, combined with a flat outline of the typical chip shown in Fig. 18b; it can be inferred that chips at this stress level may be formed by the coactions of shear failure and tensile fracture in the form of the development of Surface crack A. Figure 17c shows that the groove depth decreases further, and this decrease is confirmed by the decreased thickness in typical chips shown in Fig. 18c. It can be inferred from the above analysis that the increasing minimum confining stress may also influence the optimum spacing between cutters.

4 Discussion

Indentation efficiency is determined mainly by the volume of grooves and the consumed energy in the indentation tests. With reference to previous work (Bejari and Khademi

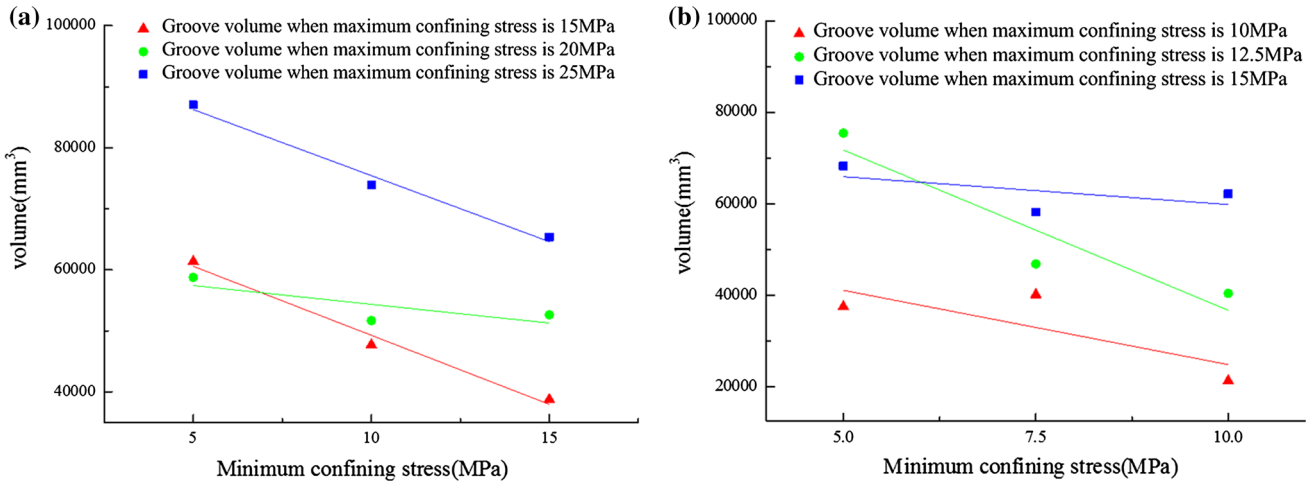


Fig. 21 Groove volumes as the minimum confining stress increases: **a** groove volumes of granite specimens; **b** groove volumes of sandstone specimens

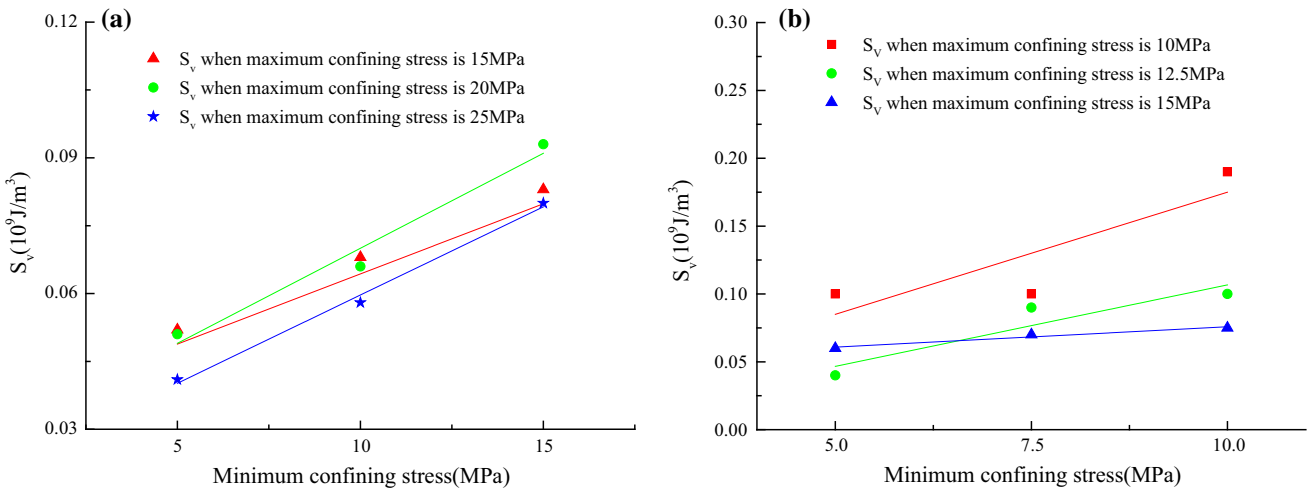


Fig. 22 S_v as the minimum confining stress increases: **a** S_v of granite specimens; **b** S_v of sandstone specimens

Hamidi 2013; Gertsch et al. 2007; Cho et al. 2013), the 'specific volume' introduced in this work can be written as:

$$\text{Specific volume } (S_v) = \frac{\text{Consumed energy}}{\text{volume of the groove}} \quad (1)$$

From the above analysis of the indentation tests, groove morphology and crack distributions, similar results of the effect of confining stress on consumed energy and groove volume can be obtained from granite and sandstone specimens. As shown in Figs. 19 and 20, increasing a minimum and maximum confining stress will increase the energy consumed in the indentation tests. Combined with the averaged maximum indentation forces in Table 2, it can be determined that as the confining stress increases, a higher thrust force is needed for TBM advancement in rocks in higher confining stress conditions. This increase in thrust force has been observed in Gehring's field investigations (Gehring 1995), and a higher consumption of TBM cutters under this condition results mainly from this increased thrust force (Gehring 1995).

Figure 21 and the above analysis of the influences of minimum confining stress on surface and internal crack propagation indicate that the volumes of the grooves formed between nicks will be decreased by the increasing minimum confining stresses. So, it can be inferred from Eq. 1 and the fitted data in Fig. 22 that S_v will increase by the increased minimum confining stress. This means that the indentation efficiency will be restrained by the increased minimum confining stress

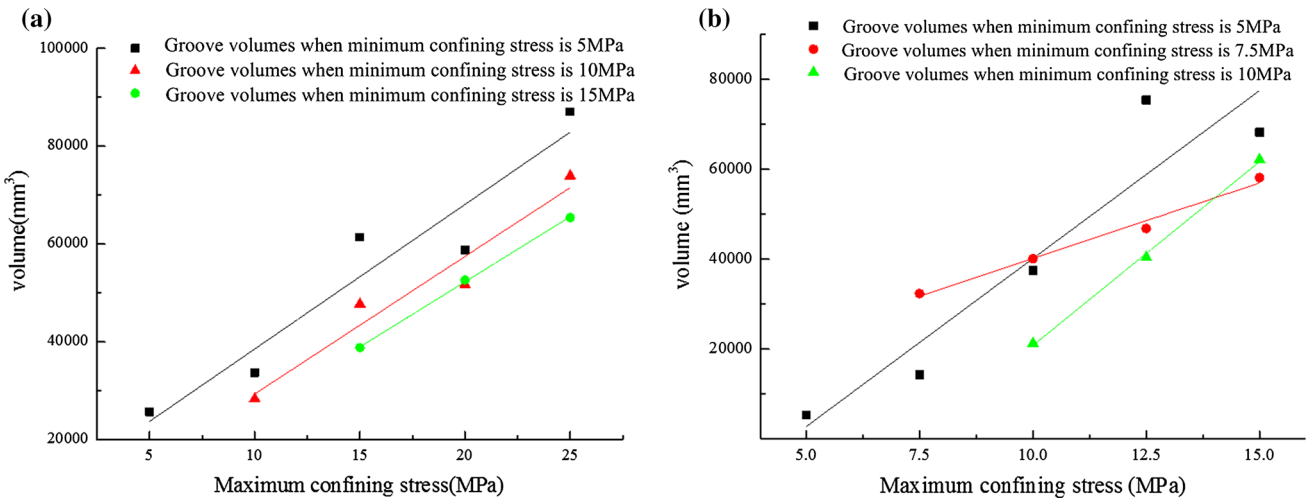


Fig. 23 Groove volumes as the maximum confining stress increases: **a** groove volumes of granite specimens; **b** groove volumes of sandstone specimens

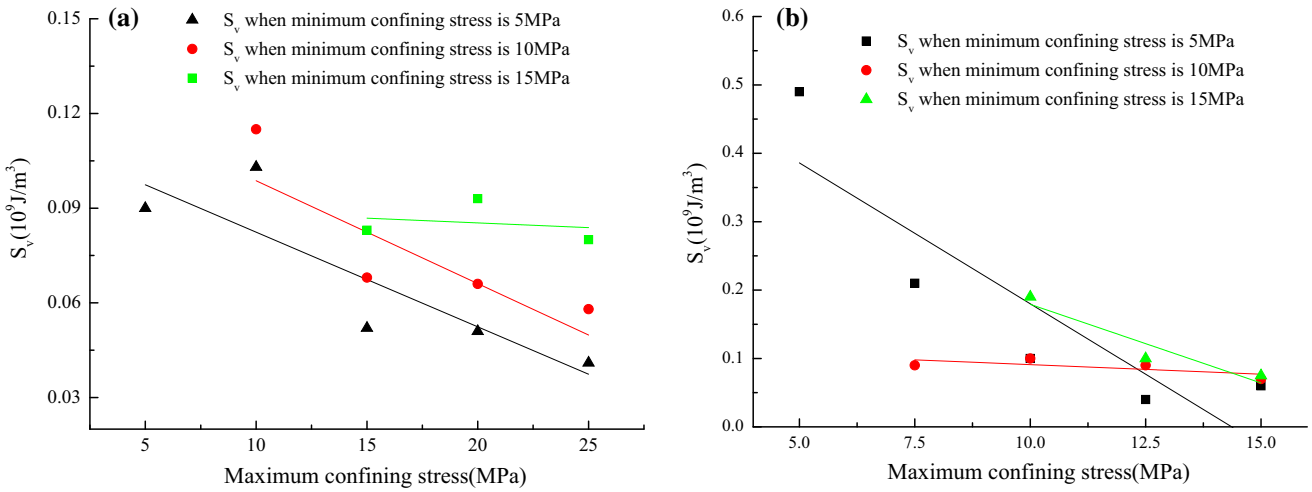


Fig. 24 S_v as the maximum confining stress increases: **a** S_v of granite specimens; **b** S_v of sandstone specimens

However, as shown in Fig. 23, the groove volume will be increased by the increasing maximum confining stress when the minimum confining stress is fixed, and the increased groove volume can be confirmed by an analysis of the effect of increasing maximum confining stress on internal crack propagation and typical chips. According to the above conclusion and Eq. 1, it appears that the effect of maximum confining stress on indentation is uncertain.

However, the fitted linear curves of the S_v in Fig. 24 show that although the slopes of these curves may vary, S_v tends to decrease with increasing maximum confining stress when the minimum confining stresses are fixed. Since the minimum confining stress is fixed, the increase in maximum confining stress is beneficial to rock breakage between nicks when the spacing and indentation depth are 70 and 6 mm, respectively.

Zhang's (2004) investigation shows that TBM cutter penetration belongs to typical quasi-static loading, and the fracture toughness of the rock subjected to a quasi-static load is almost constant. Furthermore, he proposes that the boring speed may be increased by the increasing load applied to the cutter, and is confirmed by the investigations conducted by Yin et al. (2014a, b). However, from the viewpoint of cutter wear, (Rostami et al. 2014; Gehring. 1995; Innaurato et al. 2007) state that cutter wear will be accelerated by the increased load. So, it seems that a balanced boring speed, which needs further investigation, can be obtained. In indentation tests, the normal force applied

to the TBM cutter nearly reaches 1000 kN. To keep the cutter wear low, and to ensure that the results from the indentation tests are comparable, a quasi-static indentation rate is used in this work.

In these tests, the cutter axis is parallel to the direction in which the maximum confining stress acts. To obtain a more comprehensive understanding of TBM excavation, additional tests with varying angles that are determined by cutter axis and maximum confining stress can be conducted in future work. Based on the results obtained in this work, it can be concluded that the higher difference between maximum and minimum confining stress is beneficial to rock breakage between cutters when the minimum confining stress is fixed. This is confirmed in Innaurato's investigations (Innaurato et al. 2007). To improve TBM efficiency and restrain cutter wear, some measurements to release the confinement in one direction may be conducted before the TBM cutter rotation in highly stressed tunnel surfaces. The release in confinement in one direction may be achieved by creating parallel nicks, which are vertical to the circular nicks formed by the TBM cutter on the tunnel surfaces

5 Conclusions and Future Work

Indentation tests, morphology measurements and crack observations were conducted to investigate the influence of confining stresses on rock breakage characteristics of granite and sandstone specimens. Indentation tests show that the energy consumed will be increased by an increase in maximum confining stresses. However, the increased groove volume from the increasing maximum confining stress will result in increasing indentation efficiency, whereas, as the minimum confining stress increases, the increasing consumed energy and decreasing groove volume will restrain the indentation efficiency. Finally, results on groove characteristics are confirmed by analysis of crack observations and chip formations.

The results in this article show that rock breakage between cutters is caused mainly by crack propagation, whereas the grain size is a critical factor that affects crack initiation and propagation (Chen and Labuz 2006; Brace 1961). Further investigation of the effect of grain size combined with mineral components should be conducted. Finally, additional investigations of the loading rate and cutter wear should also be conducted.

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